JONGHA KIM

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RESEARCH INTERESTS

I develop reliable and efficient multimodal large language model (MLLM) systems that retrieve relevant evidence, reason over complex long-context inputs, and coordinate specialized agents and tools. My research spans MLLM post-training, evidence and token selection, and agentic workflows for practical, scalable deployment.

EDUCATION

Korea University — Seoul, Republic of Korea
Ph.D. candidate in Computer Science and Engineering (GPA: 4.15/4.5) — Sep 2022–Aug 2028 (Expected)
Advisor: Prof. Hyunwoo J. Kim, MLV Lab

Korea University — Seoul, Republic of Korea
B.S. in Computer Science and Engineering & Statistics (Double Major) (GPA: 4.22/4.5) — Mar 2018–Aug 2022

EXPERIENCE

Adobe Research — San Jose, CA, USA
Research Scientist Intern, Video AI Lab — *May 2026–Sep 2026*

- Develop an MLLM-based agentic workflow and a scalable pseudo-labeling pipeline to improve supervision for language-guided video matting (Mentors: Joon-Young Lee, Seoung Wug Oh, David Seunghyun Yoon).

Google Cloud AI — Remote
Research Collaboration — *Dec 2024–Current*

- Conduct research on MLLM post-training [U1], retrieval-augmented generation [C4], efficient token selection [C3], and multi-agent prompt optimization [U2]. (Collaborator: Jinsung Yoon).

MLV Lab — Seoul, Republic of Korea
M.S. & Ph.D. Integrated Student — *Sep 2022–Current*

- Conduct research on reliable and efficient MLLM systems, focusing on post-training, long-context multimodal reasoning, and agentic workflows (Advisor: Hyunwoo J. Kim).

MLAI Lab — Seoul, Republic of Korea
Undergraduate Research Intern — *Jan 2021–Jul 2022*

- Developed an uncertainty-aware graph-based object detector using semantic and spatial similarities [W1] (Advisor: Sung Ju Hwang).

VoyagerX — Seoul, Republic of Korea
Intern — *Jul 2020–Jan 2021*

- Curated training data, optimized and deployed TensorFlow Lite (TFLite) on-device models for vFlat, a real-time mobile document scanner (10M+ downloads on Google Play Store).

PUBLICATIONS

C: Conference, J: Journal, W: Workshop, U: Under Review; * equal contribution

[C7] **Context Blindness in DPO: Mitigating Object Hallucination in MLLMs via Context-Calibrated Preference Optimization**
Byungoh Ko, Jinyoung Park, Jongha Kim, Jeehye Na, Jaewon Cho, Hyunwoo J. Kim
European Conference on Computer Vision (***ECCV***), 2026

[C6] **VideoSearch-R1: Iterative Video Retrieval and Reasoning via Soft Query Refinement**
Seohyun Lee*, Seoung Choi*, Dohwan Ko*, Jongha Kim, Hyunwoo J. Kim
European Conference on Computer Vision (***ECCV***), 2026

[C5] **DocPrune: Efficient Document Question Answering via Background, Question, and Comprehension-aware Token Pruning**
Joonmyung Choi, Sanghyeok Lee, Jongha Kim, Sehyung Kim, Dohwan Ko, Jihyung Kil, Hyunwoo J. Kim
In collaboration with Adobe Research
IEEE/CVF Conference on Computer Vision and Pattern Recognition (***CVPR***), 2026

[C4] **Relevance-aware Multi-context Contrastive Decoding for Retrieval-augmented Visual Question Answering**
Jongha Kim, Byungoh Ko, Jeehye Na, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with Google Cloud AI
IEEE/CVF Winter Conference on Applications of Computer Vision (***WACV***), 2026

[C3] **TabFlash: Efficient Table Understanding with Progressive Question Conditioning and Token Focusing**
Jongha Kim, Minseong Bae, Sanghyeok Lee, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with Google Cloud AI
AAAI Conference on Artificial Intelligence (***AAAI***), 2026

[C2] **VidChain: Chain-of-Tasks with Metric-based Direct Preference Optimization for Dense Video Captioning**
Ji Soo Lee*, Jongha Kim*, Jeehye Na, Jinyoung Park, Hyunwoo J. Kim
AAAI Conference on Artificial Intelligence (***AAAI***), 2025

[C1] **Groupwise Query Specialization and Quality-Aware Multi-Assignment for Transformer-based Visual Relationship Detection**
Jongha Kim*, Jihwan Park*, Jinyoung Park*, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim
IEEE/CVF Conference on Computer Vision and Pattern Recognition (***CVPR***), 2024

[J1] **Improved Query Specialization for Transformer-based Visual Relationship Detection**
Jongha Kim, Jihwan Park, Jinyoung Park, Jinyoung Kim, Sehyung Kim, Hyunwoo J. Kim
Information Sciences, 2026

[W2] **Concept Bottleneck with Visual Concept Filtering for Explainable Medical Image Classification**
Injae Kim*, Jongha Kim*, Joonmyung Choi, Hyunwoo J. Kim
MedAGI Workshop at International Conference on Medical Image Computing and Computer-Assisted Intervention (***MedAGI@MICCAI***), 2023 (**Oral Presentation**)

[W1] **Object Detection in Aerial Images with Uncertainty-Aware Graph Network**
Jongha Kim, Jinheon Baek, Sung Ju Hwang
VOLI Workshop at European Conference on Computer Vision (***VOLI@ECCV***), 2022

[U3] **SuperClip Pyramid for Video Temporal Grounding**
Sanghyeok Lee, Juyeon Ko, Joonmyung Choi, Jongha Kim, Dohwan Ko, Ji Soo Lee, Hyunwoo J. Kim
In collaboration with KT Gen AI Lab
Under Review

[U2] **CEPO: Coordinator-Executor Prompt Optimization for Multi-Agent Document Understanding**
Jongha Kim*, Seohyun Lee*, Jonathan Minho Chung, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with Google Cloud AI
Under Review

[U1] **Reason in the Words You Speak: Idiolectal Paraphrasing Off-Policy Traces for Reasoning Distillation in VideoLLMs**
Ji Soo Lee, Jinyoung Park, Seohyun Lee, Jongha Kim, Joonmyung Choi, Jinsung Yoon, Hyunwoo J. Kim
In collaboration with Google Cloud AI
Under Review

PATENTS

Method and apparatus for dense video captioning based on chained task learning and metric loss for fine-grained understanding of video language model
Hyunwoo Kim, Ji Soo Lee, Jongha Kim, Jeehye Na, Jinyoung Park
Korea Patent Application No. 10-2025-0164550.

Information providing method and system for sharing fruit information including sugar content information measured through image vision processing
Sanghoon Lee, Hyemin Song, Jongha Kim, Hyun Kim
Korea Patent Application No. 10-2020-0153010.

Method for measuring sugar content of apple using image
Sanghoon Lee, Hyemin Song, Jongha Kim
Korea Patent No. 10-2242155, registered on Apr 14, 2021.

HONORS & AWARDS

Travel Grant, CVPR 2024 — Jun 2024
Graduate School Outstanding Freshman Scholarship, Korea University — Sep 2022
Dean's List, Korea University — Spring 2019
Semester High Honors, Korea University — Fall 2018/2019/2021, Spring 2019/2020/2021
2020 Agrifood Public Big Data Startup Competition, Rural Development Administration — Aug 2020
National Scholarship for Science and Engineering, Korea Student Aid Foundation — Spring 2018–Spring 2022
Selected Trainee, 10th SW Maestro Program — May 2019–Nov 2019

ACADEMIC SERVICE

Reviewer for **NeurIPS** (2024–2026), **CVPR** (2024–2026), **ICLR** (2025–2026), **AAAI** (2025–2027), **AISTATS** (2025–2026), **ICML** (2025), **WACV** (2026–2027), and **MedAGI@MICCAI** (2024–2026)

TEACHING & TALKS

Teaching Assistant & Mentor, BoostCamp AI Tech Cohorts 1–3 — Dec 2020–Jun 2022
Upstage AI, Collaborated with Prof. Tae Hyun Oh — Seongnam, Republic of Korea

Teaching Assistant, Special Topics in Artificial Intelligence (AAA740) — Fall 2024
Korea University — Seoul, Republic of Korea

Korea University & LG AI Workshop — Feb 2023
Korea University — Seoul, Republic of Korea

Defining and Solving Problems in Deep Learning Projects — Nov 2021
BoostCamp AI Tech 2nd cohort, Upstage AI — Virtual

SKILLS

Language: Korean (native); English (fluent)
ML & Systems: Python, PyTorch, Transformers, Accelerate, DeepSpeed, vLLM, SGLang, Docker, Slurm, Git, Bash

REFERENCES

Hyunwoo J. Kim, Associate Professor, School of Computing, KAIST — *Ph.D. Advisor*
Joon-Young Lee, Principal Scientist, Adobe Research — *Internship Mentor*
Seoung Wug Oh, Senior Research Scientist, Adobe Research — *Internship Mentor*
David Seunghyun Yoon, Senior Research Scientist, Adobe Research — *Internship Mentor*
Jinsung Yoon, Staff Research Scientist and Manager, Google Cloud AI — *Research Collaborator*

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